



CSE3008 – Introduction to Machine Learning
Class Number: AP2025264000556
Slot: L33+L34
ASSIGNMENT – 3

Last Date for Submission: 13-01-2026 @ 10.00PM

Implement Linear regression and multiple linear regression on IRIS dataset.

The objective of this assignment is to:

- Implement Simple Linear Regression and Multiple Linear Regression using the IRIS dataset.
- Understand how regression behavior changes as the number of input features increases.
- Visualize regression outputs as a line, plane, and higher-dimensional hyperplane.
- Interpret model parameters and evaluate prediction performance.

Assignment Tasks:

1. Load the data set
2. Data Understanding
3. Simple Linear Regression (Line)
 - a. Predict *Petal Length* using *Petal Width*
 - i. Feature Selection - Independent variable (X): Petal Width, Dependent variable (y): Petal Length
 - ii. Train-Test Split
 - iii. Model Training
 - iv. Prediction
 - v. Visualization
 - b. Predict *Petal Length* using *Petal Width*
4. Multiple Linear Regression
 - a. Predict *Petal Length* using *Petal Width* and *Sepal Length*
5. Multiple Linear Regression (Hyperplane)
 - a. Predict Petal Length using, Petal Width, Sepal Length, Sepal Width

Assignment Submission Requirements & Constraints:

- Use NumPy, Pandas and required libraries,
- Data to be loaded from CSV / Excel
- Handwritten Code must be:
 - Clearly commented
 - Well structured
 - Accompanied by sample outputs